

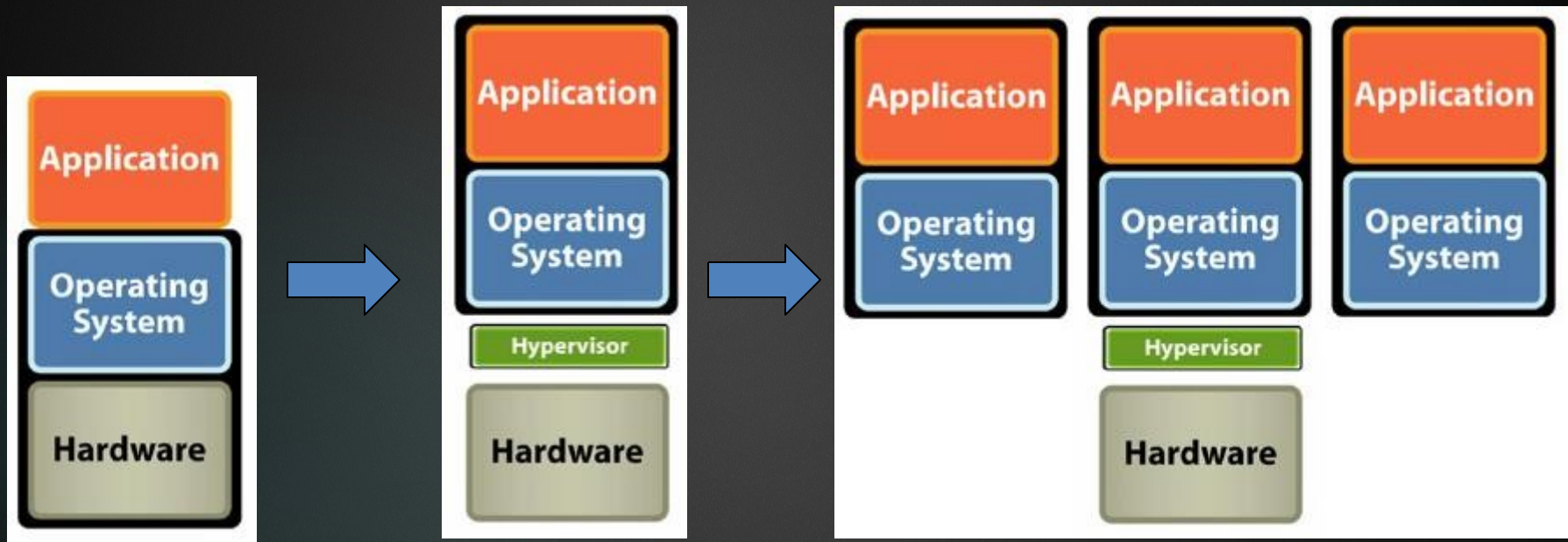
Cloud Computing

- **Virtualization**

- Virtualization, in computing, is the creation of a virtual (rather than actual) version of something, such as a hardware platform, operating system, a storage device or network resources.
- The ability to run multiple operating systems on a single physical system and share the underlying hardware resources
- Server virtualization is a technology for partitioning one physical server into multiple virtual servers. Each of these virtual servers can run its own operating system and applications, and perform as individual server.

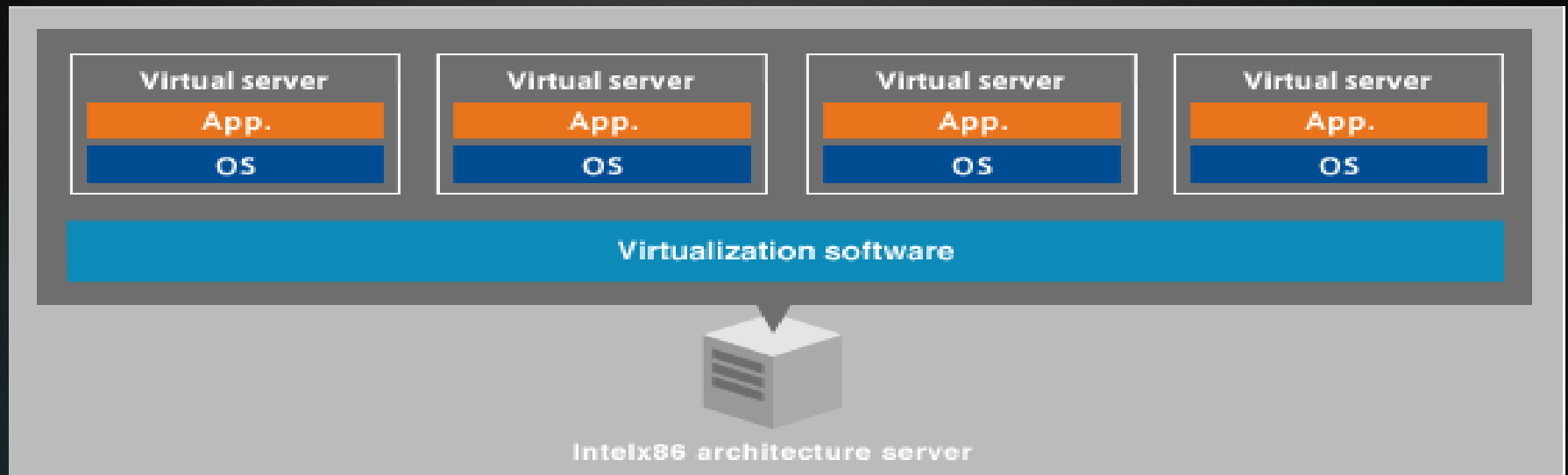
Virtualization and Cloud Computing

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Virtualization and Cloud Computing

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Terminology

- Host Machine
 - The physical hardware/server
- Hypervisor/Virtual Machine Monitor (VMM)
 - The virtualization software
 - Acts as the true 'OS' for the server
- Virtual Machines
 - Instances of the virtualized OS
 - Sometimes called Guest OS

Advantages

- Lower number of physical servers - reduce hardware maintenance costs
- Deploy multiple operating system technologies on a single hardware platform
- Easier redundancy and disaster recovery
- Reducing power consumption
- Improving server management
- Easy migration of virtual machines
- Test environments